Case Closed Final Product (Team 11)

Introduction

Mobile puzzle and detective games have become really popular lately, and it's not hard to see why. People like feeling involved in a mystery, but they don't always want long stories or complicated mechanics. A lot of existing apps try to give that investigative feeling but end up relying on guesswork or cluttered interfaces that take away from the experience instead of supporting it. Instead of helping players think through clues, these designs often leave them confused or frustrated. The real problem isn't the demand for detective games, but the lack of a focused, intuitive, and logic-driven experience.

Our target users reflect this clearly. Many of them are students squeezing in short gameplay sessions between classes or casual mobile gamers who want something fun that doesn't require a huge time commitment. Some are puzzle fans who want logical challenges rather than random tapping or unclear clues. Others just need something they can quickly save and return to later. From the personas we studied, these users aren't looking for a full-length video game. They want short, meaningful detective moments they can fit into their existing routines.

These insights led us to identify the core needs our design has to meet. Players need to be able to save and load progress easily so the experience fits into short time windows. They need a flow that works in 10–15 minutes, where they can compare suspects, check evidence, or make a verdict without feeling rushed or confused. Since most people will be using their phones, mobile-first usability is essential, like clear buttons, readable layouts, and simple navigation. And to make the detective experience feel real, immersive cues like sound effects and clear hotspots help guide the player without overwhelming them.

The purpose of the final product is to show a high-fidelity prototype of *Case Closed* that brings all these needs together. Our focus is on usability and clarity, making sure the interface actually supports investigative logic instead of getting in the way of it. The prototype demonstrates how players can move through the experience smoothly, compare information effectively, and engage with a mystery in a way that feels satisfying, even if they only have a few minutes.

Final Product Design

The final product of *Case Closed* was built entirely in VS Code using Flutter and Stack. The design uses a mobile-first approach because most of the intended players would be interacting with the app during short breaks or in between activities. The prototype supports seamless navigation between different parts of the case, including evidence, suspects, and instructions, giving users an experience that feels consistent and intuitive.

The structure of the home screen reflects this goal by presenting four different cases on the home screen, giving players an entry point into the experience. Three of the cases were not built in full, but the buttons are there to demonstrate plans for future expansion. The Suspect Comparison features are simulated through overlays and pop-ups that mirror what the interface

would show in a real implementation of building a case. These small interactions matter because they help players understand the flow of the game and how their decisions affect progress.

Core Features

Save & Load System (Figure 3)

One of the main requirements from interviews and testing was the ability to pick up where users left off. Many players did not have time for long sessions, so saving became essential. The prototype includes a save button (located in the bottom right corner), accessible from each screen during gameplay. When the user selects “Save Progress,” a confirmation overlay appears, signaling that the action has been recorded. After saving, the player returns back to the main screen where they can select a different case. For players, confirmation overlays reassures users that progress has been recorded, reducing anxiety and supporting flexible game sessions for returning players.

Suspect Comparison Tool (Figure 2)

Another core feature is the suspect comparison layout. Instead of hiding information behind complicated menus, suspects are placed vertically, allowing users to interact with each one. With this feature, users can directly compare their alibis, motives, and any connected evidence. Columns make it easy to cross-reference details and look for contradictions. This setup reflects how a detective might organize a physical case board, and it also aligns with how users naturally scan information on a mobile screen. This interaction turns the process of elimination into a satisfying and logical activity, rather than guesswork.

Evidence Viewer (Figure 4)

The evidence viewer was designed to give players control over how they investigate. Hotspots in the crime scene trigger pop-up overlays with evidence details, catered specifically to the case. From there, the player can collect the evidence. These interactions add agency to the experience, because gathering every item isn’t required to move forward, but choosing what to collect can influence how players form their theories. The dynamic overlays float above the scene rather than navigating to a new screen, keeping players immersed and focused, while visual emphasis on the hotspots ensures. Users responded positively to this feature, noting that hotspots were easy to find and that collecting evidence made them feel more like actual investigators, not just tapping through a set sequence of screens.

Interaction Design & Experience

The interaction design draws on several Human-Computer Interaction (HCI) concepts. Signifiers appear throughout the interface in the form of icons, arrows, buttons, and highflights. They tell users where to tap without requiring instructions. Mappings are clear as well: suspects are arranged vertically for easy comparison, and there’s a search suspect button that allows the player to possibly collect more information, which mirrors a true search and seizure. The conceptual model behind the design is a “digital case file.” Each tab, button, or screen reflects something that would exist in a physical investigation, from folders to photos to highlighted clues. This helps players quickly understand how to navigate, even if they have never seen the app before.

Instead of placing obvious hotspots everywhere, evidence is integrated naturally into the crime scene environment. We didn't want the game to feel too easy or like players were just tapping circles on a screen. Players must observe and reason about what seems suspicious, tapping items they consider relevant. This gives the player more control and makes the interactions feel meaningful. They are not just picking up everything they see but thinking about which clues matter to the case. This approach created a stronger sense of discovery and made the crime scene feel more alive. By encouraging careful observation of the scene rather than simple tapping, the experience mirrors the thought processes of a real investigator

After collecting clues, the player can file a warrant by selecting a suspect, The system then provides feedback based on how accurate the selection is. If the player is close but incorrect, the warrant rejection appears with a different color to indicate how far off they were. When the player selects the correct combination, the text turns green and confirms that the warrant was submitted successfully. This mechanic provides players clear visual feedback, based on the accuracy of their verdict. It also reinforced the importance of careful investigation, adding a bit of excitement to the decision-making process.

Throughout the prototype, visual feedback plays a critical role in supporting several interaction types:

- **Manipulating:** Tapping hotspots to collect evidence.
- **Exploring:** Navigating between screens and comparing suspects.
- **Conversing:** Reading feedback messages in overlay windows.
- **Responding:** Receiving visual cues on selections, such as confirmation messages or color changes.

This combination ensures that there are active participants in every step of the investigation - they observe, hypothesize, test, and draw conclusions while maintaining a satisfying user experience.

Accessibility & Usability

Accessibility was considered throughout the design. Large tap zones were used so that users could easily interact with the app without precision clicking. This matters especially for mobile contexts where players might be standing, walking, or distracted. Navigation is kept simple, with only a few major areas to explore, reducing cognitive load. The typography is clean and readable, and the layout scales smoothly for different phone sizes. Another key consideration is offline usability. Because many of the target users might be on campus or moving between places, the product is designed so it does not require Wi-Fi. This makes the app more reliable and removes barriers that could prevent people from playing.

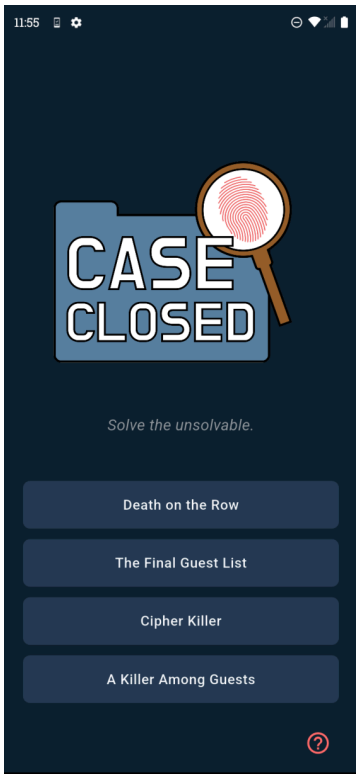
The product started with understanding what users actually needed. Early interviews showed that players wanted clarity and immersion more than complex mechanics. The personas helped make this real. Most of them only had short chunks of time to play, so being able to save progress was necessary, not optional. The personas also shaped what could reasonably happen in one session. This influenced how the screens were laid out and how quickly players could move through the case. Scenarios helped guide these decisions. For example, one

persona played during a break between classes, so the experience needed to feel complete in about 15 minutes. Because of this, navigation was designed to be fast and simple, with every screen only one or two taps away.

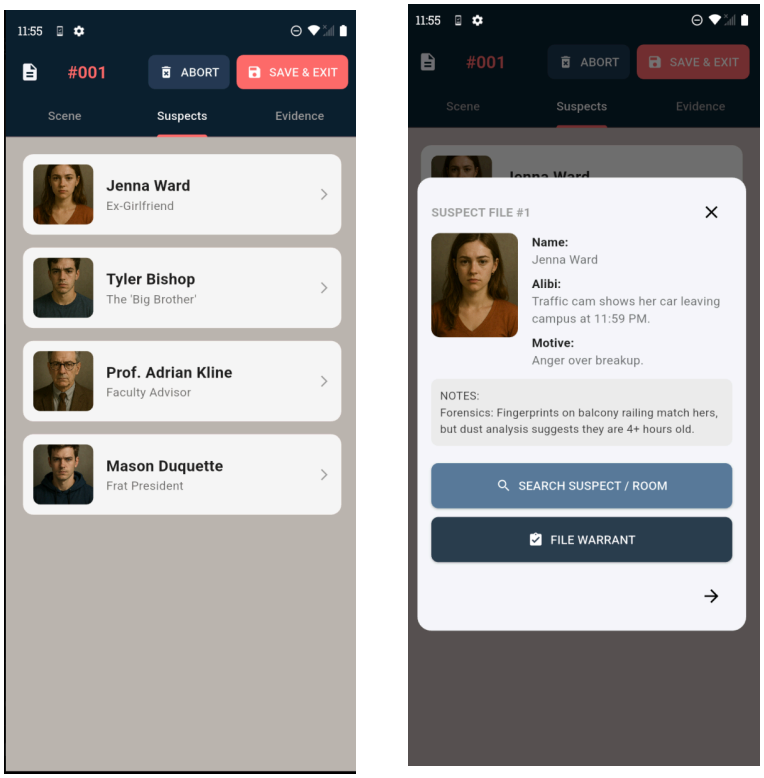
Feedback from testing shaped the layout. Some struggled with reading small handwriting in the paper prototype, so the digital version used larger fonts and clearer icons. Each iteration made the design more focused on user needs and removed anything that created confusion.

From beginning to end, the prototype evolved from rough sketches to polished digital screens. At every step, the design aimed to reflect how users actually think when solving a mystery. Instead of overwhelming them with details, the interface supports a flow where players can make quick decisions, observe patterns, and form conclusions. While it does not solve the case for them, it gives them enough structure to feel confident in their reasoning.

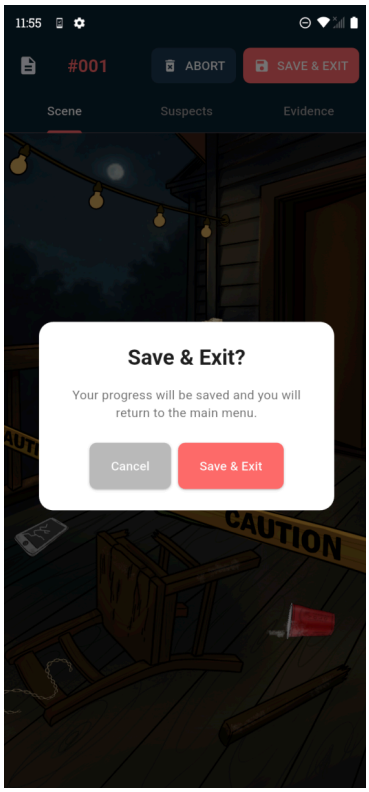
- Figure 1. Home Screen



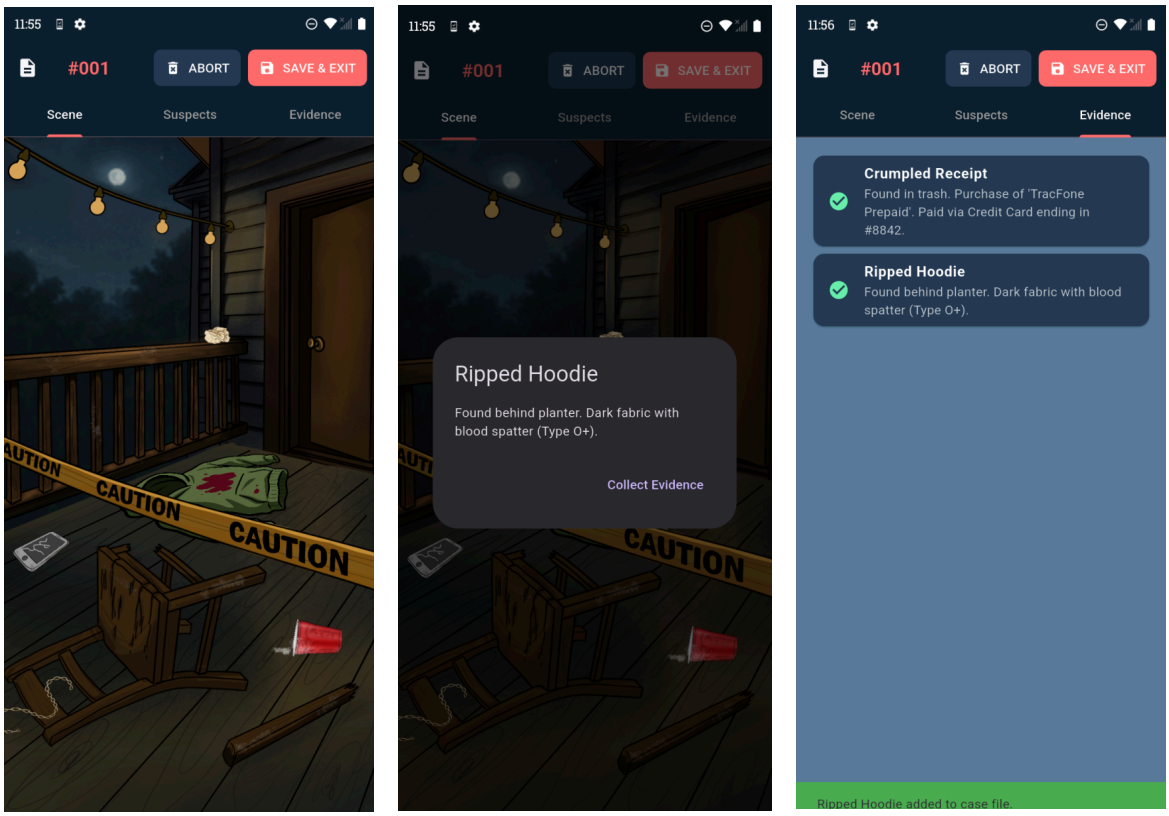
- Figure 2. Suspect Comparison



- Figure 3. Save & Load System



- Figure 4. Evidence Viewer



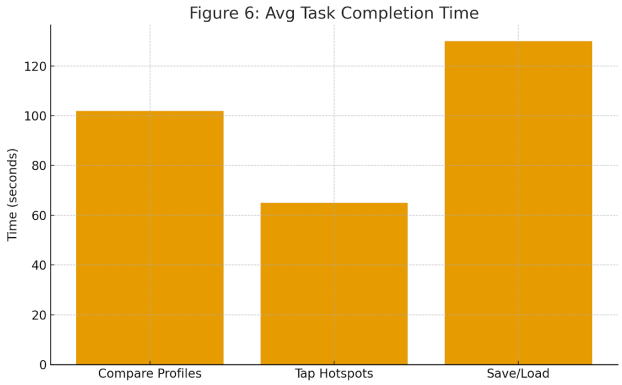
Evaluation Methodology

The prototype was evaluated using a combination of interface review and live usability testing. This approach allowed us to observe how real users interacted with the app while also checking the design against common principles such as clarity, consistency, visibility of system status, and how well the interface matched user expectations. The usability sessions helped us track how participants moved through the application, what they noticed first, and where confusion might occur during gameplay.

A total of 10 participants were recruited to represent our primary user groups: students playing between classes, casual mobile gamers, and puzzle fans who enjoy logic-based challenges. All participants provided consent and were informed of the purpose of the testing. Their demographics reflected patterns from earlier persona development, such as limited session time (10–15 minutes), quick access to information, and the need to revisit cases using save/load functionality. Unlike the earlier assumption-based design work, this stage involved direct interaction with human users, which allowed us to gather authentic feedback and observe real behaviors.

Each participant completed a series of tasks aligned with the core features of the prototype. All users were asked to compare suspect profiles and cross-check alibis. Nine out of ten participants completed this task successfully, with an average time of 1 minute 42 seconds. Tapping hotspots to locate evidence had a 100% success rate, with an average of 1 minute 05 seconds, because the interactive areas were clearly signified. The save/load task was

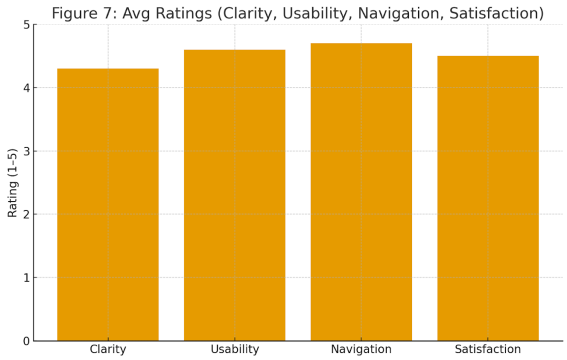
successful for nine participants, though it took slightly longer (average 2 minutes 10 seconds) due to users reading overlay messages before confirming. **Figure 6**



Participants also provided ratings for usability, clarity, and navigation on a 1–5 scale. The averages were:

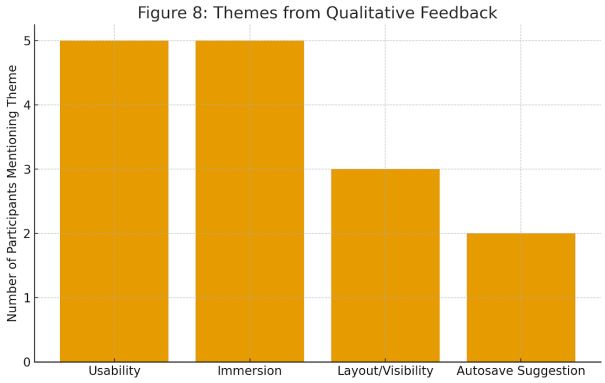
- Usability: 4.6
- Clarity: 4.3
- Navigation: 4.7

The most common negative feedback was that text and icons felt slightly difficult to read on smaller screens. Three participants mentioned this issue, which was categorized under the theme of layout and visibility.



Qualitative feedback was grouped into meaningful categories such as usability, immersion, and layout. Five participants commented that confirmation messages and overlays helped actions

feel meaningful and immersive. Two participants suggested adding an autosave feature to prevent losing progress when switching screens.



By combining numerical results with descriptive comments, the evaluation provided a clear understanding of both strengths and areas for improvement. Tasks were completed successfully, and ratings were high, but accessibility and mobile scaling were areas where small adjustments could improve the experience. Overall, the prototype demonstrated that users could navigate, discover clues, and save progress in ways that supported how they naturally think and play.

Results

Quantitative Findings

The results from testing showed that players were able to use the interface without getting confused. Average ratings were high across the board: clarity was 4.3/5, usability was 4.6/5, and navigation was 4.7/5. This means that players knew what to do and could move through the game easily. The simple layouts and clear buttons helped users find their next step without needing extra explanation.

Task performance also showed strong results. Players completed every major action in the game, including comparing suspects, finding evidence in the crime scene, and using the save/load buttons. The average time to complete each task ranged from 65 seconds to 130 seconds. The success rates were high: 100% for interacting with evidence, 87.5% for save/load, and 87.5% for suspect comparison. This suggests that most players could complete the game tasks without getting stuck.

Players also rated how confident they felt during the case, and the average score was 4.5/5. This matched the goal of creating a short but meaningful detective experience. Overall, the numbers show that players understood how the game worked and were able to solve the mystery in a short amount of time.

Qualitative Findings

The written feedback helped explain why things went well. Players liked that the save/load buttons were easy to find and that the confirmation pop-ups made them feel like their progress was safe. Many also mentioned that the comparison screen helped a lot because all the suspects were side-by-side. Being able to switch suspects on and off made it easier to see contradictions. Feedback cues, like highlighting and messages on the screen, made the interactions feel more responsive and immersive.

There were also a few things that players wanted improved. Some wanted autosave when switching screens because it matched how they used the game on their phones. A few people said that some text and icons were hard to read on smaller screens. Others wanted a way to take notes or highlight information while comparing suspects. These suggestions came up more than once and showed that players were trying to reason through the mystery, not just click through screens.

Direct User Feedback

Users often described specific parts of the experience when giving feedback. Some examples included:

> “The comparison screen made it easy to spot contradictions.”
>
> “The save button was obvious, and I liked the confirmation message.”
>
> “I wish my progress autosaved when switching screens.”
>
> “Some text felt small on my phone.”

These statements matched both the success rates and the improvement suggestions.

Overall Interpretation

Overall, the results showed that users understood the workflow and could navigate the case from beginning to end without confusion. The core detective loop, like finding clues, comparing suspects, and forming a verdict, worked as intended. Most tasks were completed quickly and successfully, and users enjoyed the sense of discovery.

Small improvements are still needed, especially around mobile readability and autosave. These changes do not affect core functionality, but they would make the experience smoother and more accessible. Even with these limitations, the prototype demonstrated that the design supports short, meaningful detective play that fits into the lives of the intended users.

Discussion

Overall, the prototype met the core needs identified from user feedback. Players could navigate easily between screens, compare suspects side-by-side, and discover evidence through

interactions in the crime scene. The mobile-first layout ensured actions felt simple and direct, which helped users move quickly and confidently through the case. Tasks were completed without confusion, and testing showed that players could finish a full session in a short amount of time. This was important because many users only had small windows to play, such as between classes or while waiting. The prototype clearly supported short play sessions, which was one of the most important requirements that came from early testing.

However, some needs were only partially met. The product did not include autosave, even though several users expected it in a mobile experience where interruptions are common. While the save/load buttons worked correctly, the lack of autosave created anxiety about losing progress if the app closed or if a notification pulled them away. This fear was mentioned more than once during testing. Another limitation was annotation. Players could not take notes or highlight clues, even though they often wanted to track their reasoning. When comparing suspects and alibis, they wanted a way to mark something that looked suspicious or come back to it later. These issues do not undermine the design, but they show where the next iteration should focus to make the experience more flexible and comfortable.

One of the main lessons learned from the evaluation is that accessibility matters more than expected in short play sessions. Clear fonts, strong contrast, and large tap zones are essential when players are using their phones quickly between activities. Even small readability problems can interrupt the flow. A few testers mentioned that some text felt too small, especially on older or smaller devices. If a player had to squint or pinch to zoom, it broke their focus and made the experience feel less smooth.

Feedback was also extremely important. Confirmation overlays and highlighting helped reinforce immersion and made players feel that their actions had consequences. These small visual cues made every interaction feel more responsive. The comparison layout was also a major success. Having suspects lined up side-by-side reduced guesswork, which was something many users disliked in other mystery games. When information was organized visually, players felt more confident making decisions. They were not just tapping randomly; they were thinking like investigators.

If the prototype were developed further, several improvements would be prioritized. First, autosave would be added so progress is never lost when switching screens or closing the app. Second, larger fonts and icons would improve readability on smaller devices. Third, a note-taking or highlighting tool would allow players to mark information, keep track of their ideas, and revisit clues. Finally, the way evidence is layered on the screen could be made more intuitive so users can see relationships between clues more easily.

These changes all align with feedback from testing and would strengthen usability without altering the core interaction model. Future designers building similar mobile mystery experiences should test early with realistic scenarios and real players. Running through the experience made it obvious where confusion might happen. Clarity should always be prioritized over visual complexity, especially on small screens, and designers should validate that players can use the game with one hand, since many play sessions will happen while walking, standing, or multitasking.

One of the most surprising findings was how much users valued immersion. Visual feedback and the feeling of uncovering clues were a major part of the enjoyment. In some cases, immersion mattered more than puzzle difficulty. Another unexpected insight was the importance of save/load. Players cared just as much about keeping their progress as solving the mystery

itself. Being able to leave and return without losing work gave the experience a level of reliability that users really appreciated.

Conclusion

This project moved from understanding user needs to building a working prototype and evaluating how well it supported detective gameplay. The process began with requirements gathered from personas, interviews, and scenarios that highlighted what players wanted: quick sessions, clear evidence, and reliable progress saving. The product included core features such as suspect comparison, hotspot discovery, and a save/load system. Evaluation showed that users could navigate smoothly and complete tasks with confidence. Quantitative results reflected high satisfaction and success rates, while qualitative feedback emphasized clarity, immersion, and ease of play.

The evaluation also revealed areas for improvement. The absence of autosave and annotation tools limited flexibility, especially in short play sessions where interruptions were common. Some users found text on smaller screens difficult to read, suggesting a need for larger fonts and icons. These insights informed recommendations for the next iteration: add autosave by default, expand accessibility through improved readability, and introduce highlighting or note-taking tools for deeper reasoning. Even with these limitations, the core detective loop worked well, and players felt engaged. The prototype demonstrated that a mobile-first mystery experience can be both intuitive and satisfying when designed around clear navigation, visual comparison, and meaningful feedback.